

[Vol. XVII., p. 60.]

3677. (Proposed by Professor Cayley.)—Find at any point of a plane curve the angle between the normal and the line drawn from the point to the centre of the chord parallel and indefinitely near to the tangent at the point; and examine whether a like question applies to a point on a surface and the indicatrix section at such point.

[Vol. XVII., p. 72.]

3564. (Proposed by Professor Cayley.)—To determine the least circle enclosing three given points.

[Vol. XVIII., July to December, 1872, p. 68.]

3875. (Proposed by Professor Cayley.)—Given the constant a and the variables x, y , to construct mechanically $\frac{a^2 - x^2}{y}$; or what is the same thing, given the fixed points A, B , and the moving point P , to mechanically connect therewith a point P' such that PP' shall be always at right angles to AB , and the point P' in the circle APB .

[Vol. XX., July to December, 1873, pp. 106, 107.]

3430. (Proposed by W. J. C. Miller.)—Find the equation of the first negative focal pedal of (1) an ellipsoid, and (2) an ellipse.

Solution by Professor Cayley.

1. It is easily seen that if a sphere be drawn, passing through the centre of the given quadric and touching it at any point (x', y', z') , then the point (x, y, z) on the required surface, which corresponds to (x', y', z') , is the extremity of the diameter of this sphere which passes through the centre of the quadric. We thus easily find the expressions

$$x = x' \left(2 - \frac{t}{a^2} \right), \quad y = y' \left(2 - \frac{t}{b^2} \right), \quad z = z' \left(2 - \frac{t}{c^2} \right);$$

where

$$t = x'^2 + y'^2 + z'^2.$$

Solving these equations for x', y', z' , and substituting in the two equations

$$xx' + yy' + zz' = x'^2 + y'^2 + z'^2, \quad \frac{x'^2}{a^2} + \frac{y'^2}{b^2} + \frac{z'^2}{c^2} = 1,$$

we get

$$\frac{x^2}{2 - \frac{t}{a^2}} + \frac{y^2}{2 - \frac{t}{b^2}} + \frac{z^2}{2 - \frac{t}{c^2}} = t, \quad (1)$$

$$\frac{x^2}{a^2 \left(2 - \frac{t}{a^2}\right)^2} + \frac{y^2}{b^2 \left(2 - \frac{t}{b^2}\right)^2} + \frac{z^2}{c^2 \left(2 - \frac{t}{c^2}\right)^2} = 1. \quad (2)$$

Since (2) is the differential with respect to t of (1), the result of eliminating t between these two equations is the discriminant of (1). Hence the equation of the required surface is the discriminant of (1) with respect to t . Since (1) is only of the fourth degree, this discriminant is easily formed. If (1) be written in the form

$$At^4 + 4Bt^3 + 6Ct^2 + 4Dt + E = 0,$$

it will be found that A and B do not contain x, y, z , while C, D, E contain them, each in the second degree. Now the discriminant is of the sixth degree in the coefficients, and of the form $A\phi + B^2\psi$ (see Salmon's *Higher Algebra*, §107); hence it contains x, y, z only in the tenth degree. This is therefore the degree of the required surface.

The section of this derived surface by the principal plane z consists of the discriminant of

$$\frac{x^2}{2 - \frac{t}{a^2}} + \frac{y^2}{2 - \frac{t}{b^2}} = t, \quad (3)$$

(which is of the sixth degree, and is the first negative pedal of $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$), together with the conic (taken twice), which is obtained by putting $t = 2c^2$ in (3).

This conic, which is a double curve on the surface, touches the curve of the sixth degree in four points.

2. The formulae for the conic are quite analogous to those for the ellipsoid, viz. we have

$$x = X \left\{ 2 - \frac{1}{a^2}(X^2 + Y^2) \right\}, \quad y = Y \left\{ 2 - \frac{1}{b^2}(X^2 + Y^2) \right\},$$

leading to the equations

$$\theta = \frac{x^2}{2 - \frac{\theta}{a^2}} + \frac{y^2}{2 - \frac{\theta}{b^2}},$$

and its derived equation, from which to eliminate θ . The first is the cubic equation

$$(A, B, C, D)(\theta, 1)^3 = 0,$$

where

$$A = 1, \quad B = -\frac{2}{3}(a^2 + b^2), \quad C = \frac{1}{3}(a^2x^2 + b^2y^2 + 4a^2b^2), \quad D = -2a^2b^2(x^2 + y^2).$$

Equating the discriminant to zero, this is

$$0 = A^2 \nabla = 4(AC - B^2)^3 - (3ABC - A^2D - 2B^3)^2.$$

Or finally

$$(3a^2x^2 + 3b^2y^2 - 4a^4 + 4a^2b^2 - 4b^4)^3 \\ + \{9(a^2 - 2b^2)a^2x^2 + 9(b^2 - 2a^2)b^2y^2 - 8a^6 + 12a^4b^2 + 12a^2b^4 - 8b^6\}^2 = 0,$$

which is the required equation.

[Vol. XXI., January to June, 1874, pp. 29, 30.]

4298. (Proposed by J. W. L. Glaisher, B.A.)—With four given straight lines to form a quadrilateral inscribable in a circle.

Solution by Professor Cayley.

Let the sides of the quadrilateral taken in order be a, b, c, d ; and let its diagonals be x, y ; viz. x the diagonal joining the intersection of the sides a, b with that of the sides c, d ; y the diagonal joining the intersection of the sides a, d with that of the sides b, c ; then, the quadrilateral being inscribed in a circle, the opposite angles are supplementary to each other. Suppose for a moment that the angles subtended by the diagonal x are $\theta, \pi - \theta$, we have

$$x^2 = b^2 + c^2 + 2bc \cos \theta, \quad x^2 = a^2 + d^2 - 2ad \cos \theta;$$

and thence

$$(ad + bc)x^2 = ad(b^2 + c^2) + bc(a^2 + d^2) = (ac + bd)(ab + cd),$$

that is,

$$x^2 = (ac + bd) \frac{ab + cd}{ad + bc},$$

and similarly

$$y^2 = (ac + bd) \frac{ad + bc}{ab + cd},$$

agreeing as they should do with the known relation $xy = ac + bd$: the quadrilateral is thus determined by means of either of its diagonals. It is however interesting to treat the question in a different manner.

Considering a, b, c, d, x, y as the sides and diagonals of a quadrilateral, we have between these quantities a given relation, say

$$F(a, b, c, d, x, y) = 0,$$

and the quadrilateral being inscribed in a circle, we have also the relation $xy = ac + bd$; which two equations determine x, y ; and thus give the solution of the problem.

The expression of the function F is in effect given in my paper, "Note on the value of certain determinants, &c.," *Quarterly Mathematical Journal*, t. iii. (1860), pp. 275–277, [286]; viz. a, b, c being the edges of any face, and f, g, h the remaining edges of a tetrahedron, then

$$\begin{aligned} \text{volume} = \frac{1}{144} \{ & b^2 c^2 (g^2 + h^2) + c^2 a^2 (h^2 + f^2) + a^2 b^2 (f^2 + g^2) \\ & + g^2 h^2 (b^2 + c^2) + h^2 f^2 (c^2 + a^2) + f^2 g^2 (a^2 + b^2) \\ & - a^2 f^2 (a^2 + f^2) - b^2 g^2 (b^2 + g^2) - c^2 h^2 (c^2 + h^2) \\ & - a^2 g^2 h^2 - b^2 h^2 f^2 - c^2 f^2 g^2 - a^2 b^2 c^2 \}, \end{aligned}$$

where, when the tetrahedron becomes a quadrilateral, the volume is $= 0$.

In this formula, changing c, b, h, g, f, a into a, b, c, d, x, y , we have the required equation $F = 0$; viz. this is found to be

$$\begin{aligned} & a^2 b^2 c^2 + b^2 c^2 d^2 + c^2 d^2 a^2 + d^2 a^2 b^2 - b^2 d^2 (b^2 + d^2) - a^2 c^2 (a^2 + c^2) \\ & + x^2 y^2 (a^2 + b^2 + c^2 + d^2 - x^2 - y^2) \\ & + x^2 (a^2 c^2 + b^2 d^2 - a^2 d^2 - b^2 c^2) y^2 (a^2 c^2 + b^2 d^2 - a^2 b^2 - c^2 d^2) = 0, \end{aligned}$$

which, with $xy = ac + bd$, determines x, y . Substituting in the foregoing equation for xy its value, the equation becomes

$$(ad + bc)^2 x^2 + (ab + cd)^2 y^2 = 2\{a^2 b^2 c^2 + b^2 c^2 d^2 + c^2 d^2 a^2 + d^2 a^2 b^2 + abcd(a^2 + b^2 + c^2 + d^2)\},$$

or

$$(ad + bc)^2 x^2 + (ab + cd)^2 y^2 = 2(ad + bc)(ab + cd)(ac + bd).$$

To show more clearly how this equation arises, I observe that we have identically

$$\begin{aligned} F - (a^2 + b^2 + c^2 + d^2 - x^2 - y^2)(xy + ac + bd)(xy - ac - bd) \\ - 2(ad + bc)(ab + cd)(xy - ac - bd) = \{(ad + bc)x - (ab + cd)y\}^2. \end{aligned}$$

The resulting equation $(ad + bc)x - (ab + cd)y = 0$, together with $xy = ac + bd$, gives for x, y the foregoing values.

[Vol. XXI., pp. 81, 82.]

4392. (Proposed by S. Roberts, M.A.)—If N_p denotes the number of terms in a symmetrical determinant of p rows and columns, show that the successive numbers are given by the equation

$$N_k - N_{k-1} - (k-1)^2 N_{k-2} + \frac{1}{2}(k-1)(k-2)\{N_{k-3} + (k-3)N_{k-4}\} = 0,$$

k being positive and N_0 being taken equal to unity.

Solution by Professor Cayley.

It is a curious coincidence that the question of determining the number of distinct terms in a symmetrical determinant has been recently solved by Captain Allan

Cunningham in a paper in the last number of the *Quarterly Journal of Science*¹; and the question having been proposed to me by Mr Glaisher, I have also solved it in a paper [580] printed in the April Number of the *Monthly Notices of the Royal Astronomical Society*. I there obtain

$$N_k = 1.2 \dots k \text{ coeff. } x^k \text{ in } \frac{e^{x+\frac{1}{2}x^2}}{(1-x)^{\frac{1}{2}}},$$

viz. writing

$$u = N_0 + N_1 \frac{x}{1} + N_2 \frac{x^2}{1.2} + \dots,$$

I show that u satisfies the differential equation

$$2 \frac{du}{dx} = \left\{ 1 + x + \frac{1}{1-x} \right\} u,$$

giving when the constant is determined

$$u = \frac{e^{x+\frac{1}{2}x^2}}{(1-x)^{\frac{1}{2}}}.$$

Writing the differential equation in the form

$$2(1-x) \frac{du}{dx} = (2-x^2)u,$$

we at once obtain for N_k the equation of differences

$$N_k - kN_{k-1} + \frac{1}{2}(k-1)(k-2)N_{k-3} = 0,$$

which is in fact a particular first integral of Mr Roberts's equation; viz. from the above equation we have

$$N_{k-1} - (k-1)N_{k-2} + \frac{1}{2}(k-2)(k-3)N_{k-4} = 0,$$

and multiplying this last by $k-1$ and adding, we have

$$N_k - N_{k-1} - (k-1)^2 N_{k-2} + \frac{1}{2}(k-1)(k-2)\{N_{k-3} + (k-3)N_{k-4}\} = 0,$$

which is the equation obtained by Mr Roberts. It thence appears that the general first integral of his equation is

$$N_k - kN_{k-1} + \frac{1}{2}(k-1)(k-2)N_{k-3} = (-)^k C 1.2 \dots (k-1).$$

The equation

$$N_k = kN_{k-1} - \frac{1}{2}(k-1)(k-2)N_{k-3}$$

gives very readily the numerical values, viz.

$$\begin{array}{lll} 1 = 1.1 - 0, & 17 = 4.5 - 3.1, & 2461 = 7.388 - 15.17, \\ 2 = 2.1 - 0, & 73 = 5.17 - 6.2, & 18155 = 8.2461 - 21.73, \\ 5 = 3.2 - 1.1, & 388 = 6.73 - 10.5. & \end{array}$$

[Vol. XXII., July to December, 1874, pp. 20, 21.]

4354. (Proposed by R. Tucker, M.A.)—Solve the equations

$$-x^2 + xy + xz = a = 4, \tag{1}$$

$$-y^2 + xy + yz = b = -20, \tag{2}$$

$$-z^2 + xz + yz = c = -8. \tag{3}$$

¹I have not the volume at hand to refer to, but he obtains an equation of differences, and gives the numbers 1, 2, 5, 73, 398 (should be 388),...

Note on Question 4354. By Professor Cayley.

A question of simple algebra such as this, becomes more interesting when interpreted geometrically: thus, writing the equations in the form

$$-x^2 + xy + xz = aw^2, \quad yx - y^2 + yz = bw^2, \quad zx + zy - z^2 = cw^2,$$

and then putting for shortness

$$\alpha = -a + b + c, \quad \beta = a - b + c, \quad \gamma = a + b - c,$$

the solutions obtained are

$$x : y : z : w = a\alpha : b\beta : c\gamma : (\alpha\beta\gamma)^{\frac{1}{2}},$$

$$x : y : z : w = a\alpha : b\beta : c\gamma : -(\alpha\beta\gamma)^{\frac{1}{2}};$$

say these are

$$\{\alpha\alpha, b\beta, c\gamma, (\alpha\beta\gamma)^{\frac{1}{2}}\} \quad \text{and} \quad \{a\alpha, b\beta, c\gamma, -(\alpha\beta\gamma)^{\frac{1}{2}}\}.$$

But the equations are also satisfied by

$$(x = 0, y = z, w = 0), \quad (y = 0, z = x, w = 0), \quad (z = 0, x = y, w = 0),$$

or what is the same thing, $(0, 1, 1, 0)$, $(1, 0, 1, 0)$, $(1, 1, 0, 0)$. The three equations represent quadric surfaces, each two of them intersecting in a proper quadric curve, and the three having in common 8 points; viz. these are made up of the first mentioned two points each once, and the last mentioned three points each twice: $2 + 3 \cdot 2 = 8$.

To verify this, observe that, at each of the three points, the tangent planes of the surfaces have a common line of intersection; this line is the tangent of the curve of intersection of any two of the surfaces, and the curve of intersection therefore touches the third surface; wherefore the point counts for two intersections. In fact, taking (X, Y, Z, W) as current coordinates, the equations of the tangent planes at the point (x, y, z, w) are

$$\begin{aligned} X(2x - y - z) - Yx - Zx + 2aWw &= 0, \\ -Xy + Y(-x + 2y - z) - Zy + 2bWw &= 0, \\ -Xz - Yz + Z(-x - y + 2z) + 2cWw &= 0 : \end{aligned}$$

hence at the point $(0, 1, 1, 0)$ these equations are

$$-2X = 0, \quad X + Y - Z = 0, \quad -X - Y + Z = 0,$$

which three planes meet in the line $X = 0, Y - Z = 0$; and similarly for the other two of the three points.

[Vol. XXII., pp. 60-64.]

4458. (Proposed by Professor Cayley.)—Find (1) the intersections of the two quartic curves

$$\lambda(ab - xy)^2 = abx(a - y)(b - y), \quad \mu(ab - xy)^2 = aby(a - x)(b - x);$$

and (2) trace the curves in some particular cases; for instance, when $a = 1$, $b = 2$, $\lambda = 1$, $\mu = -2$.

Solution by the Proposer.

1. The 16 intersections are made up as follows: 5 points at infinity on the line $x = 0$, 5 at infinity on the line $y = 0$, the two points $(x = a, y = b)$, $(x = b, y = a)$, and 4 other points, $16 = 5 + 5 + 2 + 4$. As to the points at infinity, observe that, as regards the first curve, the point at infinity on the line $x = 0$ is a flecnode having this line for a tangent to the flecnodal branch; and, as regards the second curve, the same point is a cusp, having this line for its tangent; hence the point in question counts as $2 + 3 = 5$ intersections; and the like as to the point at infinity on the line $y = 0$. It remains to find the coordinates of the 4 points of intersection. Assume $xy = ab\omega$, then the equations become

$$\lambda(1 - \omega)^2 = x + \omega y - (a + b)\omega, \quad \mu(1 - \omega)^2 = \omega x + y - (a + b)\omega;$$

hence, eliminating successively y and x , the factor $1 - \omega$ divides out,—this factor belongs to the points $(x = a, y = b)$, $(x = b, y = a)$ for which obviously $\omega = 1$ —, and the equations become

$$(\lambda - \mu\omega)(1 - \omega) + (a + b)\omega = (1 + \omega)x,$$

$$(\mu - \lambda\omega)(1 - \omega) + (a + b)\omega = (1 + \omega)y.$$

Multiplying these two equations together, and substituting for xy its value $ab\omega$, we find

$$\{(\lambda - \mu\omega)(\mu - \lambda\omega) + (a + b)(\lambda + \mu)\omega\}(1 - \omega)^2 + (a + b)^2\omega^2 - (1 + \omega)^2\omega ab = 0.$$

Write, for shortness, $p = (\lambda + \mu)(a + b) - \lambda^2 - \mu^2$, then, dividing by ω^2 , and writing $\omega + \frac{1}{\omega} = \Omega$, the equation is

$$(\lambda\mu\Omega + p)(\Omega - 2) + (a + b)^2 - ab(\Omega + 2) = 0;$$

viz. this is a quadric equation for Ω . But, instead of Ω , it is convenient to introduce the quantity $\theta, = \frac{\Omega - 2}{\Omega + 2} = \left(\frac{\omega - 1}{\omega + 1}\right)^2$. The equation thus becomes

$$\left\{2\lambda\mu\frac{1 + \theta}{1 - \theta} + p\right\}\frac{4\theta}{1 - \theta} + (a + b)^2 - ab\frac{4}{1 - \theta} = 0,$$

or

$$\{2\lambda\mu(1 + \theta) + p(1 - \theta)\}4\theta + (a + b)^2(1 - \theta)^2 - 4ab(1 - \theta) = 0,$$

or

$$\theta^2\{(a + b)^2 - 4(p - 2\lambda\mu)\} + \theta\{-2a^2 - 2b^2 + 4(p + 2\lambda\mu)\} + (a - b)^2 = 0;$$

viz. substituting for p its values, this is

$$\theta^2(a+b-2\lambda-2\mu)^2 + 2\theta\{-a^2 - b^2 + 2(\lambda+\mu)(a+b) - 2(\lambda-\mu)^2\} + (a-b)^2 = 0;$$

or if we write

$$A = a^2 - 2a(\lambda+\mu) + (\lambda-\mu)^2, \quad B = b^2 - 2b(\lambda+\mu) + (\lambda-\mu)^2,$$

this is

$$\theta^2(a+b-2\lambda-2\mu)^2 - 2(A+B)\theta + (a-b)^2 = 0,$$

whence

$$\begin{aligned} \{(a-b)^2 - (A+B)\theta\}^2 &= \theta^2\{(A+B)^2 - (a-b)^2(a+b-2\lambda-2\mu)^2\} \\ &= \theta^2\{(A+B)^2 - (A-B)^2\} = 4AB\theta^2; \end{aligned}$$

viz. taking for convenience the sign $-$ on the right-hand side, this is

$$(a-b)^2 - (A+B)\theta = -2\theta\sqrt{AB};$$

and we have thus

$$\theta = \frac{(a-b)^2}{(\sqrt{A}-\sqrt{B})^2},$$

that is,

$$\theta = \frac{\omega-1}{\omega+1} = \frac{a-b}{\sqrt{A}-\sqrt{B}}; \quad \omega = \frac{\sqrt{A}-\sqrt{B}+a-b}{\sqrt{A}-\sqrt{B}-a+b}.$$

We may write

$$\begin{aligned} x &= \mu(\omega-1) + \frac{1}{2}(a+b) + \frac{1}{2}(a+b-2\lambda-2\mu)\frac{\omega-1}{\omega+1}, \\ y &= \lambda(\omega-1) + \frac{1}{2}(a+b) + \frac{1}{2}(a+b-2\lambda-2\mu)\frac{\omega-1}{\omega+1}; \end{aligned}$$

whence also $x-y = (\mu-\lambda)(\omega-1)$, as is also seen at once from the original equations; then we have

$$\begin{aligned} \frac{1}{2}(a+b-2\lambda-2\mu)\frac{\omega-1}{\omega+1} &= \frac{\frac{1}{2}(a-b)(a+b-2\lambda-2\mu)}{\sqrt{A}-\sqrt{B}} \\ &= \frac{\frac{1}{2}(A-B)}{\sqrt{A}-\sqrt{B}} = \frac{1}{2}(\sqrt{A}+\sqrt{B}); \end{aligned}$$

and the values are

$$\begin{aligned} x &= \frac{2\mu(a-b)}{\sqrt{A}-\sqrt{B}-a+b} + \frac{1}{2}(\sqrt{A}+\sqrt{B}+a+b) \\ &= \frac{(a-b)(\mu-\lambda)+b\sqrt{A}-a\sqrt{B}}{\sqrt{A}-\sqrt{B}-a+b}, \\ y &= \frac{2\lambda(a-b)}{\sqrt{A}-\sqrt{B}-a+b} + \frac{1}{2}(\sqrt{A}+\sqrt{B}+a+b) \\ &= \frac{(a-b)(\lambda-\mu)+b\sqrt{A}-a\sqrt{B}}{\sqrt{A}-\sqrt{B}-a+b}, \end{aligned}$$

which may be expressed in the more simple form

$$\begin{aligned} x &= \frac{1}{4\lambda}(a+\lambda-\mu+\sqrt{A})(b+\lambda-\mu+\sqrt{B}), \\ y &= \frac{1}{4\mu}(a-\lambda+\mu+\sqrt{A})(b-\lambda+\mu+\sqrt{B}), \end{aligned}$$

the transformations depending on the identity

$$\frac{8\lambda\mu(a-b)}{\sqrt{A}-\sqrt{B}-a+b} = ab - (\lambda + \mu)(a+b) + (\lambda - \mu)^2 \\ + \sqrt{A}(b - \lambda - \mu) + \sqrt{B}(a - \lambda - \mu) + \sqrt{AB},$$

which is easily verified. Of course, since the signs of $\sqrt{A}$, $\sqrt{B}$ are arbitrary, we have 4 systems of values of (x, y) , which is right.

In the original equations, for a, b, λ, μ, x, y , write $1, k^{-2}, \lambda^2, -\mu^2, x^2, -y^2$; then the equations become

$$\lambda^2(1 + k^2x^2y^2)^2 = x^2(1 + y^2)(1 + k^2y^2), \\ \mu^2(1 + k^2x^2y^2)^2 = y^2(1 - x^2)(1 - k^2x^2),$$

and we thence have

$$\lambda + \mu i = \frac{x\sqrt{(1+y^2)(1+k^2y^2)} + iy\sqrt{(1-x^2)(1-k^2x^2)}}{1 + k^2x^2y^2};$$

viz. assuming $x = \operatorname{sn} \alpha$ ($\operatorname{sinam} \alpha$), $iy = \operatorname{sn} i\beta$, this is $\lambda + \mu i = \operatorname{sn}(\alpha + \beta i)$; viz. the problem is (for a given modulus k , assumed as usual to be real, positive, and less than 1) to reduce a given imaginary quantity $\lambda + \mu i$ to the form $\operatorname{sn}(\alpha + \beta i)$. The proper solution is that in which the signs of the radicals are each $-$, viz. it may in this case be shown that the value of x^2 is positive and less than 1, that of y^2 positive. The values thus are

$$x^2 = \frac{1}{4\lambda^2}(1 + \lambda^2 + \mu^2 - \sqrt{A})\left(\frac{1}{k^2} + \lambda^2 + \mu^2 - \sqrt{B}\right), \\ y^2 = \frac{1}{4\mu^2}(1 - \lambda^2 - \mu^2 - \sqrt{A})\left(\frac{1}{k^2} - \lambda^2 - \mu^2 - \sqrt{B}\right),$$

where

$$A = 1 - 2\lambda^2 + 2\mu^2 + (\lambda^2 + \mu^2)^2, \quad B = \frac{1}{k^2} - \frac{2}{k^2}\lambda^2 + \frac{2}{k^2}\mu^2 + (\lambda^2 + \mu^2)^2.$$

The solution is really equivalent to that given by Richelot (*Crelle*, t. XLV., 1853, p. 225). To verify this partially, observe that, writing σ, τ for Richelot's $\tan \frac{1}{2}\phi$, $\tan \frac{1}{2}\psi$, we have

$$\left(\sigma + \frac{1}{\sigma}\right)\lambda = 1 + \lambda^2 + \mu^2,$$

giving

$$\left(\sigma - \frac{1}{\sigma}\right)\lambda = -\sqrt{A}; \\ \left(\tau + \frac{1}{\tau}\right)\frac{\lambda}{k} = \frac{1}{k^2} + \lambda^2 + \mu^2,$$

giving

$$\left(\tau - \frac{1}{\tau}\right)\frac{\lambda}{k} = -\sqrt{B};$$

whence

$$2\sigma\lambda = 1 + \lambda^2 + \mu^2 - \sqrt{A}, \quad 2\tau\frac{\lambda}{k} = \frac{1}{k^2} + \lambda^2 + \mu^2 - \sqrt{B},$$

or the above value of x^2 is $= k^{-1}\sigma\tau$, agreeing with his; the value of y^2 is, however, presented under a somewhat different form.

2. The curves are

$$(2 - xy)^2 = 2x(1 - y)(2 - y), \quad -(2 - xy)^2 = y(1 - x)(2 - x) \quad (1, 2),$$

each passing through the points (1, 2) and (2, 1); the four points of intersection found by the foregoing general theory are all real, viz. these are

$$\begin{array}{llll} x = \frac{1}{2}(2 + \sqrt{3})(5 + \sqrt{17}), & y = -\frac{1}{8}(-1 + \sqrt{3})(-1 + \sqrt{17}), & \text{say } +17 \cdot 00 & -0 \cdot 57, \\ & -\sqrt{3}, & +\sqrt{17}, & +1 \cdot 65 \quad +0 \cdot 94, \\ & +\sqrt{3}, & -\sqrt{17}, & +1 \cdot 22 \quad +2 \cdot 13, \\ & -\sqrt{3}, & -\sqrt{17}, & -0 \cdot 12 \quad -3 \cdot 49. \end{array}$$

The equation of the curve (1) may also be written in the forms

$$y^2(x^2 - 2x) + 2yx - 4x + 4 = 0, \quad x^2y^2 + x(-2y^2 + 2y - 4) + 4 = 0.$$

The original form shows that, if y is between 1 and 2, x is negative—but by a further examination it appears that there is not in fact any branch of the curve between these limits of y —but y being outside these limits, then x is positive; in fact, the whole curve lies on the positive side of the axis of y . And then the inspection of the first quadric equation shows that the lines $x = 0$ and $x = 2$ are each an asymptote.

The point at infinity on the axis of y is in fact a flecnodal branch, the tangent to the flecnodal branch being $x = 0$, and that of the ordinary branch $x = 2$.

Similarly, from the second quadric equation, it appears that the line $y = 0$ is an asymptote; the point at infinity on the axis of x is in fact a cusp, the axis in question $y = 0$ being the cuspidal tangent.

The equation of the curve (2) may also be written in the forms

$$x^2y^2 + (x^2 - 7x + 2)y + 4 = 0, \quad (y^2 + y)x^2 - 7yx + 2y + 4 = 0.$$

The original form shows that, if x is between 1 and 2, y is positive; but that x being beyond these limits, y is negative; and as regards the first case, x between 1 and 2, we at once establish the existence of an oval, meeting the line $y = 1$ in the points $x = 2$ and $\frac{3}{2}$, and the line $y = 2$ in the points $x = 1$ and $\frac{4}{5}$; it is further easy to see that the horizontal tangents of the oval are $y = \frac{1}{10}(25 \pm \sqrt{113})$, = say $2 \cdot 2$ and $0 \cdot 9$.

The remainder of the curve lies wholly below the line $y = 0$. The first quadric equation shows the asymptote $x = 0$; the point at infinity on the axis of y is in fact a cusp, having the axis itself for the cuspidal tangent. The second quadric equation shows the asymptotes $y = 0$, $y = -1$; the point at infinity on the axis of x is in fact a flecnodal branch, having the line $y = 0$ for the tangent to the flecnodal branch, and $y = -1$ for that of the other branch. It is further seen that there are two vertical tangents $x = \frac{1}{2}(11 \pm \sqrt{113}) = 10 \cdot 8$ or $0 \cdot 2$; the former of these touches a branch